

DAX PATEL

Embedded Software Engineer | Embedded Linux | System Software

Ahmedabad, India | +91 90544 20861 | daxbpatel22@gmail.com | linkedin.com/in/daxpatel06

SUMMARY

Embedded software engineer with hands-on expertise in C, Linux kernel customization, Device Tree configuration, driver integration, and Yocto Project builds. Proven track record in hardware/software co-design and system bring-up on an AMD ZCU102 (Zynq UltraScale+ MPSoC) integrated with a custom RISC-V subsystem. Strong foundation in Electronics & Communication Engineering, bridging the gap between low-level boot flow, driver integration, and system-level debugging on real hardware.

TECHNICAL SKILLS

Languages	C, 8051 Assembly
Embedded Linux	Linux Kernel Configuration, Device Tree, Driver Integration & Porting, Yocto Project (BitBake), Xilinx OSL Flow, DRM/KMS (modetest), Cross-compilation, IPC, TCP/IP
Platforms & Tools	AMD/Xilinx ZCU102, Zynq UltraScale+ MPSoC, RISC-V, ARM Cortex-A53, AXI, DDR, Vitis, Vivado, XSCT/JTAG, Git
Hardware	Microcontroller peripherals (GPIO, LCD, keypad, timers), Analog circuit design, PCB design and hardware bring-up, Multisim, Proteus
Multimedia (Beginner)	GStreamer, Linux multimedia pipeline fundamentals

EXPERIENCE

MosChip — Intern, System Software

Dec 2025 – Present

Ahmedabad, India

- Completed structured C, Linux kernel internals, device driver models, and BSP training (Dec 2025 – Feb 2026).
- Executed the Linux display pipeline bring-up project on AMD ZCU102 (Feb 2026 – Jun 2026).
- Transitioned to the Multimedia team (Jun 2026 – Present), focusing on GStreamer and multimedia pipeline integration.

PROJECTS

Display Bring-Up on AMD ZCU102 — Custom RISC-V Subsystem

Feb 2026 – Jun 2026

MosChip

End-to-end bring-up of a DDR-to-HDMI display pipeline under Linux on an AMD ZCU102 (Zynq UltraScale+ MPSoC). Validated Linux on the ARM Cortex-A53 processing system before targeting the display path to a custom RISC-V processor subsystem implemented in the FPGA programmable logic.

- Achieved Linux boot validation on the ARM Cortex-A53 using dual-boot topologies (SD card and JTAG+TFTP).
- Configured and generated custom kernel images using both the Yocto Project and Xilinx Out-of-Tree (OSL) build environments.
- Programmed FPGA bitstreams to instantiate the RISC-V soft-core and integrated the underlying RISC-V Linux software stack (OpenSBI, U-Boot, Kernel, Rootfs).
- Diagnosed non-functional display pipelines; integrated required Linux driver modifications and authored Device Tree nodes for video IP block registration.
- Debugged system-level boot and kernel panic logs to find root causes, validating the entire path with DRM/modetest to achieve active HDMI color-bar output on physical hardware.

- Authored technical documentation and build scripts to establish reproducible BSP and kernel compilation procedures.

Tools & Technologies: Linux Kernel, Device Tree, Linux Driver Integration, Yocto Project, Xilinx OSL Flow, U-Boot/OpenSBI, DRM/modetest, Cross-compilation, RISC-V, ARM Cortex-A53, AXI, Vitis, XSCT/JTAG

8051 Microcontroller Development Board (Plug-and-Play)

Academic Project

Designed and assembled a reusable 8051 development board from scratch, writing bare-metal Assembly firmware for its basic peripherals to verify hardware integrity.

- Architected and soldered the development board, selecting and wiring peripherals for modular reuse across projects.
- Authored low-level Assembly drivers for Port 1 GPIO LED control and 16x2 LCD interfacing, handling initialization sequences, command/data writes, and precise timing delays.
- Implemented 4x4 matrix keypad scanning routines featuring software debouncing and key-release detection logic.
- Configured hardware Timer0 registers directly in Assembly to generate a precise 10 Hz square wave.

Tools & Technologies: 8051 Assembly, Bare-metal Firmware, GPIO Control, LCD Interfacing, Keypad Scanning, Timer Programming, Hardware Bring-up

Variable Power Supply with Battery Backup

Academic Project

Designed a 12V power supply featuring LM317-based adjustable output voltage and an automatic relay-based switchover to a battery backup during mains failure.

- Conducted schematic capture and circuit simulation for the mains-operated variable supply stage in Multisim and Proteus.
- Prototyped and tested the rectifier, filter, and LM317 regulation stages on a breadboard before migrating to PCB.
- Designed and integrated a relay-based automatic battery backup and charging system.
- Debugged hardware faults during bring-up, resolving relay configuration issues, a faulty ammeter, and LM317 pinout routing errors on the PCB.
- Validated final system performance: achieved a smooth 1.19–13.3V adjustable output with seamless relay transition to a 12V battery on mains failure.

Tools & Technologies: Analog Circuit Design, LM317 Voltage Regulation, Rectifier/Filter Design, Relay-based Switching, PCB Design, Multisim, Proteus

EDUCATION

Dharmsinh Desai University, Nadiad — B.Tech, Electronics & Communication Engineering

May 2026

CGPA: 8.47 / 10

CERTIFICATIONS

- Digital Design with Verilog — NPTEL
- Digital System Design and Verification using CPLD Board (Krypton) — NPTEL